

Modeling NFL Field Goal Success Probability as a Function of Distance

Luke Desnoyer

Table of Contents

1. Introduction

- Research Question / Aim
- Rationale / Personal Engagement

2. Background & Data

- Overview of Field Goals in the NFL
- Data Sources and Variables Collected
- Table of Field Goal Attempts

3. Data Analysis

- Empirical Success Rates
- Graphs / Observed Patterns

4. Mathematical Modeling

- Choice of Model (Logistic Regression)
- Parameter Estimation
- Model Validation

5. Discussion & Reflection

- Assumptions and Limitations
- Summary of Findings

6. Works Cited

Introduction

i. Research Question / Aim

The aim of this exploration is to examine how the probability of a successful NFL field goal varies with distance from the goalposts by developing a mathematical model to predict success at different ranges.

ii. Rationale / Personal Engagement

I have followed football for years, and I have always been interested in the situations where a team decides whether to trust its kicker or keep the offense on the field. These decisions feel intuitive when watching a game, but I wanted to see what the numbers truly support. This project allowed me to combine an interest in sports with mathematical tools that model real-world relationships, especially in situations where outcomes are binary and probabilities shift gradually.

Background & Data

i. Overview of Field Goals in the NFL

Field goals are one of the primary scoring methods in American football, worth three points each. They are typically attempted when a team is within kicking range but unable to score a touchdown, or when time or field position makes a touchdown attempt less feasible. Field goals often occur in scenarios such as the final plays of a half, on fourth down when a team is too far for a touchdown, or during closely contested games where every point matters.

While many factors can influence field goal success, distance is by far the most significant and consistent predictor. Shorter kicks are almost always successful, while the probability of making longer kicks decreases sharply. Other variables such as kicker skill, environmental conditions, or game pressure can have some effect, but these factors are less predictable and often introduce noise rather than systematic trends.

For the purpose of this investigation, I will focus exclusively on distance as the primary variable. I have chosen not to consider weather, indoor versus outdoor stadiums, individual kicker differences, or other situational factors. This allows the analysis to remain mathematically clear while capturing the dominant influence on success rates.

ii. Data Sources and Variables Collected

To collect the necessary data, I turned to pro-football-reference.com, a comprehensive online database for American football statistics, history, and records. I began with a full table of field goal attempts, which listed the total number of attempts (FGA) and successful kicks (FGM) accross several distance. The original table contained unnecessary information such as extra-point attempts and overall field goal percentage, but those were not required for my model. Since my investigation focuses strictly on the effect of distance on field goal success, I extracted only the columns corresponding to field goal attempts and made them for each distance range. Then, I used Google Sheets to reorganize the data into a simplified dataset including only the distance bin, the number of attempts, and the number of successful kicks. This clean version is the dataset used for the subsequent calculations and modeling.

iii. Table of Field Goal Attempts

Distance Bin (Yards) ▾	FGA ▾	FGM ▾
0--19	2	2
20--29	236	229
30--39	297	280
40--49	301	231
50--59	264	191
60--69	15	4

Data plotted on Google Sheets

Note: A field goal over 69 yards was not attempted in the 2024 season. As such, the last bin is 60-69 yards.

Data Analysis

i. Empirical Success Rate

To quantify how distance affects field goal success, I calculated the empirical success rate for each distance bin. For a given bin, the success rate is the ratio of successful attempts to total attempts:

$$\text{Success Rate}_{\text{Bin}} = \frac{\text{Made FGs in Bin}}{\text{Attempts in Bin}}$$

Example calculation (29 yard bin): $229/236 = \sim 0.97$

This equation yielded the following success rate values for each bin:

Distance Bin (Yards)	FGA	FGM	Emp. Rate
0--19	2	2	1.00
20--29	236	229	0.97
30--39	297	280	0.94
40--49	301	231	0.77
50--59	264	191	0.72
60--69	15	4	0.27

Data plotted on Google Sheets

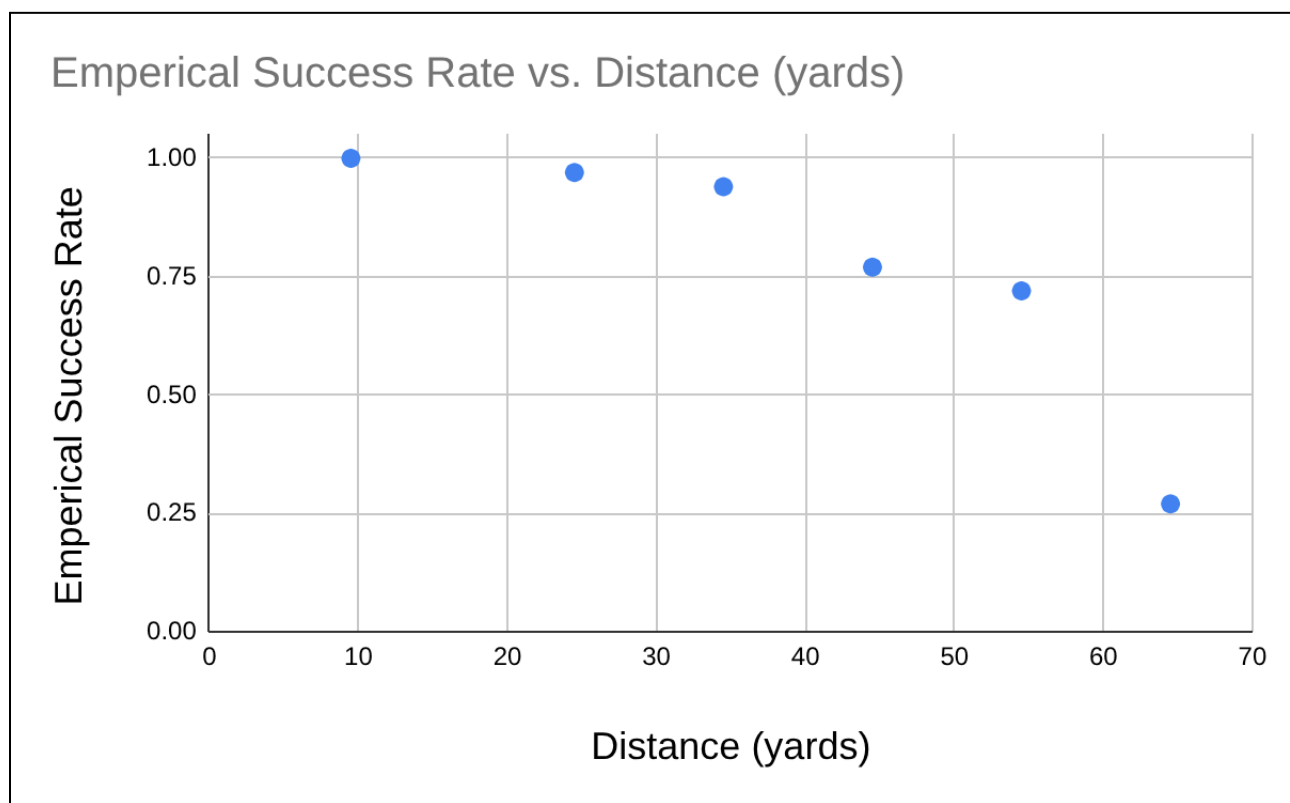
ii. Graphs / Observed Patterns

Since the field goal data was organized in distance ranges rather than exact yard lines, I picked the midpoint of each range to stand in for the whole group. It's a common way to handle grouped data, and it allowed me to plot the points cleanly so I could see how the success probability shifts as the distance increases. To convert each

distance range into a single usable value, I calculated the midpoint of the interval. For example, for the 40–49 yard bin, the midpoint is:

$$(40 + 49)/2 = 44.5$$

I applied the same process to all other bins and used these midpoint values when plotting the empirical success rates. Then, using Google Sheets, I created the following scatterplot:



Emperical Success Rate vs. Distance (yards)

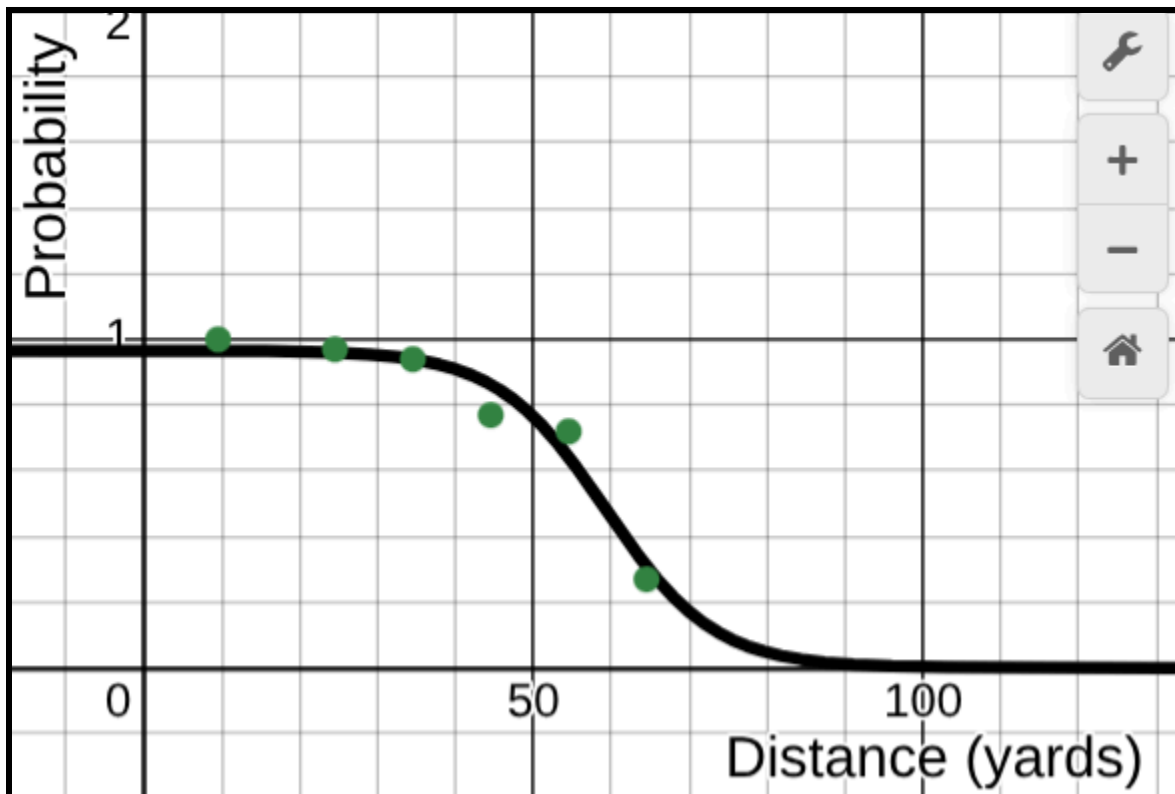
This scatterplot confirms the decreasing trend of field goal attempts: shorter kicks have a much higher probability of success, while longer kicks become increasingly difficult. What stood out to me though, was that the decline is not linear. The success rate drops slowly for short and mid ranged kicks, then falls more sharply for distances beyond 44 yards. This creates an S-shaped pattern, with the steepest decline occurring in the 50-64 yard range. I also noticed that the longest-distanced bin (60-69 yards) shows greater variability, likely because many professional kickers have an accurate range of about 55 yards and aren't used to kicking field goals over 60 yards. Overall, the

scatter plot reveals a smooth, bounded decrease in probability, which suggests that a logistic model can be an appropriate choice for representing the relationship between distance and field goal success.

Mathematical Modeling

i. Choice of Model (Logistic Regression)

Seeing the S-shaped pattern in the scatterplot made it clear that a linear model wouldn't be the best for expected point calculation. The probability starts near one, drops steadily, and levels off near zero, so I needed a function that stays within these bounds. Logistic regression fits naturally here because a field goal is either made or missed, and the model captures the curved decline in probability. Now that I had determined the model to use, I entered my data into the Desmos Graphing Calculator program to estimate the logistic function, creating a smooth curve that shows how success probability changes with distance and allows me to calculate expected points for different kicks.



Empirical Success Rate vs. Distance (yards)

This function provides a basic estimate of the probability of making a field goal at any given distance; however, several aspects of it do not reflect the realities of NFL football. To make the model more applicable to real-world scenarios, I introduced bounds based on the physical and regulatory constraints of the game. Determining the lower bound was straightforward: NFL kickers typically kick from 7 yards behind the line of scrimmage, and the end zone extends 10 yards beyond the goal line, so the minimum field goal distance is approximately 17 yards. Consequently, I set $x \leq 17$ as the minimum value on my updated graph. Establishing the maximum distance, however, requires more involved calculations.

To estimate the maximum possible distance at which an NFL kicker could physically clear the uprights (ignoring drag conditions due to wind), I modeled the football's flight as a projectile launched with initial speed v_0 , launch angle θ , and initial height h . Since the ball must pass above the 3.05 meter high crossbar rather than land on the ground, I used both the horizontal and vertical displacement equations and combined them into a single condition involving the required height.

The horizontal motion is described by

$$x = v_0 \cos\theta t$$

which allows me to express time in terms of horizontal distance,

$$t = \frac{x}{v_0 \cos\theta}.$$

The vertical motion is described by

$$y(t) = h + v_0 \sin\theta t - \frac{1}{2}gt^2.$$

Substituting the expression for t from the horizontal equation gives a height function in terms of horizontal distance:

$$y(x) = h + x \tan\theta - \frac{gx^2}{2v_0^2 \cos^2\theta}$$

A field goal is successful only if the ball is still above the crossbar height when it reaches the uprights, so I used the requirement

$$y(x) \geq 3.05$$

as a mathematical condition for a successful long kick. Solving the equation

$$h + x \tan\theta - \frac{gx^2}{2v_0^2 \cos^2\theta} = 3.05$$

gives the maximum distance at which the ball can still clear the bar.

Next, I determined the value of the constants using the output of elite NFL kickers at peak biological performance. Professional measurements suggest that elite kickers strike the ball at a maximum of approximately 122.5 km/h (~34.03 m/s) at an optimal angle of 38° during long field goal attempts (Matan K, 2023). A professional measurement of the initial height of the launch was significantly harder to find. Based on the anatomy of the average NFL kicker, kicker posture, and ball contact based on rough location data, I was able to make a reasonable assumption that the ball leaves contact with the kicker's foot at about 0.25 m above the ground.

Using this information, I was able to set $h = 0.25$, $\theta = 38^\circ$, $v_0 = 34.03$, and $g = 9.81$ (gravitational constant).

$$0.25 + x \tan(38) - \frac{9.81x^2}{2(34.03^2) \cos^2(38)} = 3.05$$

When evaluated with these constants, the equation simplifies to the following quadratic equation:

$$0.00682x^2 - 0.7813x + 2.80 = 0.$$

I was then able to solve the equation using the quadratic formula,

$$x = \frac{-(-0.7813) \pm \sqrt{(0.7813)^2 - 4(0.00682)(2.8)}}{2(0.00682)}$$

which yields two answers: $x = 3.7035$ and $x = 110.85662$.

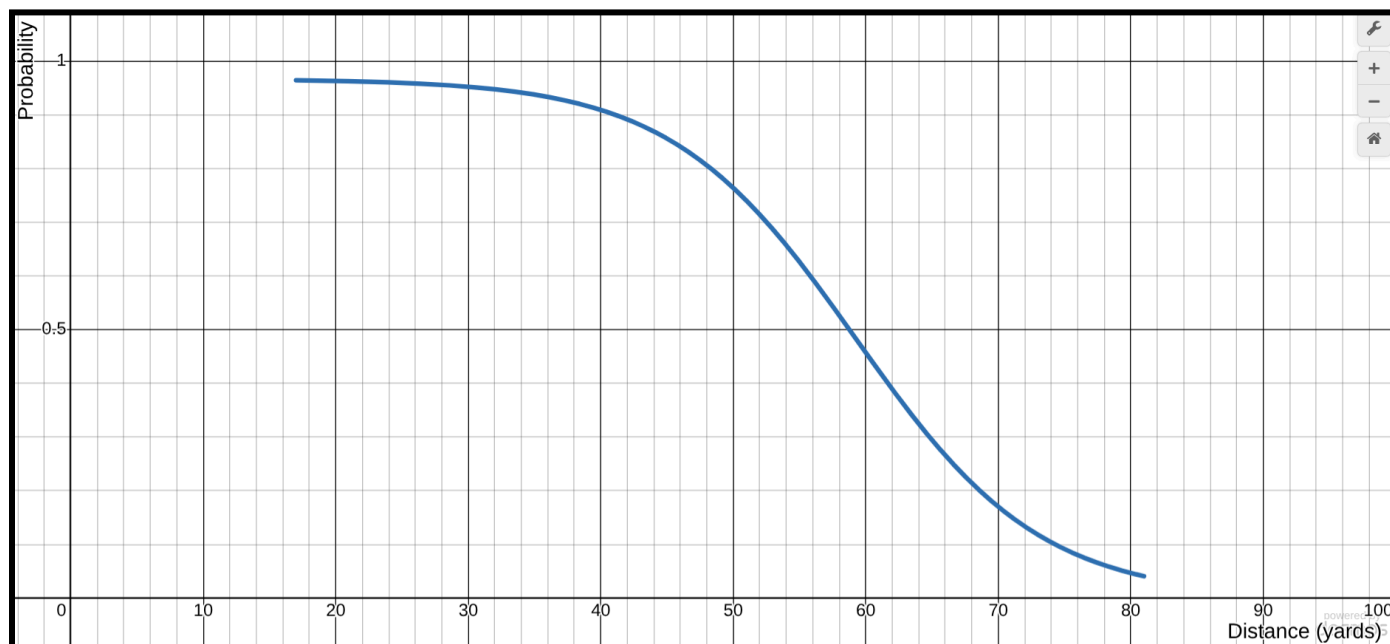
In this context, we want to know how many meters the ball has traveled when it reaches the crossbar on the way down, meaning that $x = 110.85662$ was our applicable meter value. Since my model used yards as the primary distance unit, I converted 110.86 to yards by multiplying by 1.094. This yielded a value of 121.28. Therefore, I was able to set the upper bound of my logistical model to $x = 121$ (*yards*)

Although this estimation is mathematically sound, it stood out as an unreasonable maximum distance based on the fact that no professional kicker has come close to kicking a field goal over 100 yards, even in perfect conditions. With this in mind, I realized that I would need to consider air resistance to make the upper bound of my model more accurate. A full analysis of the effect of air resistance would require solving differential equations that account for forces in both horizontal and vertical directions. However, this approach would make the model unnecessarily complex and only slightly more accurate, since even the advanced model would not account for wind, ball rotation, or other environmental pressures, all of which are conditions that go beyond my model's purpose of determining how the chance of making field goals changes at different points on the field..

Instead, I applied a simplified method. Experimental and theoretical analyses of footballs show that drag reduces a football's range by approximately 33 percent (Hartschuh). To account for this in a manageable way, I multiplied the previously calculated upper bound by a correction factor of 0.67. The factor is smaller than the maximum observed reduction in conditions, ensuring that the adjusted estimate remains realistic.

$$\text{Adjusted upper bound} = 0.67 \times \text{Original upper bound} \approx 0.67 \times 121 \text{ yds} \approx 81 \text{ yds}$$

This approach gives me a final upper bound value for my model with a reasonable degree of accuracy. The updated logistic regression curve can be seen below:



Empirical Success Rate vs. Distance (yards)

Desmos Graphing Calculator, Desmos, Inc., <https://www.desmos.com/calculator>

ii. Parameter Estimation

With the upper bound established, the next step is to set the parameters that allow the logistic equation to match the observed data. In the logistic model,

$$y = \frac{a}{1 + e^{-(bx+c)}}$$

the parameter a represents the maximum value the function can reach, showing the upper limit of the system. The parameter b controls how quickly the values grow, determining the steepness of the curve. The parameter c shifts the curve along the x-axis, indicating the point where growth is fastest.

Desmos makes estimating these parameters straightforward through its regression feature. By inputting the distance and success rate data, Desmos Graphing Calculator automatically adjusts a , b , and c to create the curve that best fits the observed points. It does this by minimizing the differences between the predicted success probabilities

from the logistic function and the actual success rates for each distance. The resulting values can then be interpreted in context: a (0.966843) shows the maximum probability a field goal can achieve, b (-0.143466) reflects how rapidly the success rate declines as distance increases, and c (8.50011) indicates the distance at which the probability is halfway to its maximum. This automated fitting process allows the model to capture the general trend of the data while smoothing out variability in the observed success rates.

EQUATION 

$$y = \frac{0.966843}{1 + e^{-(-0.143466x + 8.50011)}}$$

STATISTICS

$$R^2 = 0.9532$$

RESIDUALS

$$e_1$$

Parameters from Desmo Graphing Calculator (Logistic Regression feature)

iii. Model Validation

To validate the logistic model, I compared the predicted probabilities from the regression curve to the actual success rates observed in the data. The model fits the data very closely, as indicated by an R-squared value of 0.9532. This high coefficient of determination means that over 95% of the variation in field goal success rates is explained by the distance-based logistic model, demonstrating that it reliably captures the underlying relationship.

In addition to this validation, I wanted to compare my model to the actual output of a single NFL kicker. For this analysis, I chose Brandon Aubrey, the NFL kicker with the most field attempts during the 2024 season (ESPN Media). Using his individual statistics from the season, I created the table below comparing his outputs with my model.

Distance Range	Avg Predicted Success Rate	Actual Success Rate
0–19	0.96	No Attempts
20–29	0.95-0.96	1
30–39	0.92-0.96	0.8
40–49	0.79-0.91	0.88
50–59	0.49-0.76	0.69
60–69	0.19-0.46	0.5

Data plotted on Google Sheets

The table compares Aubrey’s actual success rate by distance range with the probabilities predicted in the logistic model. In most intervals, the observed success rate falls within the predicted range, indicating that the model generalizes well to an individual kicker. For instance, in the 40–49 yard range, Aubrey’s success rate of 0.88 lies within the model’s predicted interval of 0.79–0.91. At longer distances, particularly beyond 50 yards, he attempts and converts field goals at ranges that are uncommon league-wide, suggesting that he has exceptional leg strength compared to the average kicker in my 2024 sample. At shorter distances in the 30–39 yard range, Aubrey misses more kicks than predicted, suggesting his accuracy may be below the 2024 league average. Based on the model, Aubrey appears to be an exceptionally strong kicker who experiences reduced accuracy at certain distances, which corroborates reports from ESPN indicating a similar pattern in his performance.

Discussion & Reflection

i. Assumptions and Limitations

Despite being based in sound mathematical logic, my model is not without several flaws and limitations. Analyzing these shortcomings are important to the integrity of the analysis and for future improvements to my model.

First, my regression was based on binned data, which allowed for straightforward data collection at the cost of a small degree of accuracy. Initially, I had set out to use point data from every individual kick of the 2024 season. However, collecting that data would've required an in-depth analysis of 272 games of play-by-play reports, a feat that I estimate to take over 20 hours, which was unrealistic a figure for just the data collection portion of the analysis.

Next, my model only used data from the 2024 NFL season, making it questionably accurate when applied to other NFL seasons. With that said, kicking trends change very little by individual seasons (2023 kicking data is noticeably similar to 2024 data). Because of this, it is plausible to apply this model for the next few seasons of NFL football, assuming there are no major changes to kicking procedure.

Third, my calculations for the upper bound of my logistic graph were limited by the complex nature of football projectile physics. Standard equations of motion do not account for the unique drag experienced by a football's unique shape and rotation. Because of this, I used a correction factor to make my upper bound realistic. In a more ideal circumstance, I would consider a deeper motion analysis of a football to find a more accurate upper bound.

Finally, despite being addressed in the introduction of this analysis, it is necessary to again underscore the omission of environmental and situational factors in this model. The original plan did not include these variables because the goal has always been to determine NFL field goal success probability strictly as a function of distance. In a more comprehensive model of overall field goal success probability, additional factors would ideally be considered, including atmospheric conditions, defensive pressure, game pressure, and execution by the long snapper and holder.

ii. Summary of Findings

Overall, through the analysis of 2024 NFL kicking data, I met my goal of creating a mathematically based model of field goal success as a function of distance. The model I created clearly demonstrates the sharp decline in success rates that NFL kickers face at

distances beyond 50 yards. I validated my model mathematically using the R-squared value and practically through comparison with individual kicker statistics from the 2024 season, demonstrating that it's not only fairly accurate but also applicable to real NFL scenarios. In terms of practicality, this means that my model could be used by NFL staff to determine whether it's mathematically better to kick a field goal or go for it on 4th down based on their own team's analytics.

The usefulness of complex models based on data has spiked modern professional football as organizations start to realize the usefulness of simulation and modeling in seeking the optimal outcomes of high-stakes games. As complex modelling becomes easier with the rise of AI, I can see a future where models like mine make a majority of decisions during a football game. Whether this is detrimental to the spirit of the game or not is up to individual interpretation.

Works Cited

Pro Football Reference. "2024 NFL Kicking and Punting Statistics."

[Pro-Football-Reference.com](https://www.pro-football-reference.com), Sports Reference LLC, 2024. Accessed 24 Nov. 2025.

Finio, Ben. "Field Goal! The Science Behind a Perfect Football Kick." Science Buddies, 20 Nov. 2020,

https://www.sciencebuddies.org/science-fair-projects/project-ideas/Sports_p059/sports-science/science-behind-a-perfect-football-kick. Accessed 8 Dec. 2025.

Matan, K. "Ball-Tracking Data and the Anatomy of a Field Goal." Medium, 23 Oct. 2023,

www.medium.com/@matan_k/ball-tracking-data-and-the-anatomy-of-a-field-goal-0c646073fbe8 Accessed 9 Dec. 2025.

"What Is a Projectile?" The Physics Classroom, The Physics Classroom, n.d.,

www.physicsclassroom.com/class/vectors/lesson-2/what-is-a-projectile Accessed 9 Dec. 2025.

Hartschuh, Ryan D. "The Physics of Football." Real-World Physics Problems,

real-world-physics-problems.com/physics-of-football.html. Accessed 11 Dec. 2025.

"NFL Player Kicking Stats 2024." ESPN, ESPN Internet Ventures, 2024,

https://www.espn.com/nfl/stats/player/_/view/special/stat/kicking/season/2024/seasontype/2. Accessed 5 Jan. 2026.